Which Depth Estimator for Jetson AGX Orin?

A 2026 Benchmark of Monocular and Stereo Methods

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Abstract

The Jetson AGX Orin is a common computer for robots and vehicles, yet measured accuracy, speed and energy of current depth estimators on it are hard to find. We benchmark nine model families as of October 2026: the monocular Depth Anything V2 and 3, Depth Pro, MoGe-2, Metric3D v2 and UniDepth V2, and the stereo methods libSGM, NVIDIA ESS and Fast-FoundationStereo. Every method runs on the Orin as a TensorRT FP16 engine or native CUDA code at up to three input sizes, up to the full 5.7 MP, and is scored for metric accuracy against Middlebury 2014 ground truth (non-occluded pixels of the 10 scenes whose disparity stays below 256 px), GPU time, energy and licence. Stereo is far more accurate. At 30 fps, NVIDIA's ESS reaches 0.70% AbsRel in 8.6 ms and libSGM 0.73% in 25 ms; with more compute, Fast-FoundationStereo reaches 0.33%. The best monocular models reach 12% with the true focal length (Depth Anything 3 Metric-L in 33 ms, Depth Pro in 803 ms), and 3.8–4.0% only with an oracle per-image scale (MoGe-2-L, UniDepth V2-B, Metric3D v2-L). Stereo error falls as the input resolution grows; no monocular model improves, and the metric scale of most drifts away from the truth outside their training resolution. At depth discontinuities, which AbsRel barely registers, the monocular models scatter 17–46% of the edge points between foreground and background, learned stereo 2–9% and libSGM at full resolution 1%. We give recommendations per frame-rate budget and release all code, per-scene results and engine build scripts.

1 Introduction

Dense metric depth is a core input for driving, robotics and industrial automation, and the NVIDIA Jetson AGX Orin is one of the most common computers on which it runs. Two families of camera-based methods compete for it. *Monocular* depth networks such as Depth Anything V2 [1] and 3 [2], Depth Pro [3], MoGe-2 [4], Metric3D v2 [5] and UniDepth V2 [6] infer metric depth from a single image with large vision transformers. *Stereo* methods triangulate depth from two calibrated cameras, with classical matching such as semi-global matching (SGM) [7] or with learned matchers such as NVIDIA's ESS [8] and Fast-FoundationStereo [9].

Choosing among them for an Orin is hard. Papers report accuracy on desktop GPUs, at the networks' training resolution and often after aligning the prediction to the ground truth; embedded timings, if any, come from different devices, precisions and input sizes; and licences differ from permissive to non-commercial. This paper is a practical benchmark for that choice. We run every method on the same Jetson AGX Orin 64GB as a TensorRT FP16 engine (accuracy from FP32 where FP16 is unreliable) or as native CUDA code, at up to three input sizes from its native or smallest input to the full 5.7 MP, and score every output against the same full-resolution Middlebury 2014 ground truth [10] on the 10 scenes whose disparity stays within the 256-level range of the semi-global matcher. We report metric accuracy, GPU time, energy and licence, and derive recommendations per frame-rate budget.

Contributions. (i) Measured metric accuracy, GPU time and energy on Jetson AGX Orin for nine current monocular and stereo model families, each at up to three input sizes (table 7). (ii) Recommendations: the most accurate stereo and monocular configuration for each frame-rate budget, with licences (section 5). (iii) A resolution-scaling study: stereo error falls with resolution, monocular error does not, because the monocular metric scale drifts outside the training resolution and the shape does not improve (section 6.1). (iv) An analysis of depth discontinuities, where AbsRel is blind to the "flying" points that the networks produce and a classical matcher can be the better choice (section 6.3). (v) All code, per-scene results, engine build scripts and this paper's source, released at `https://github.com/nodarhub/orin-depth-benchmark`.

2 Related work

Monocular depth. MiDaS [11] and DPT [12] established zero-shot relative depth by mixing datasets. Depth Anything [13, 1] scales this with DINOv2 features [14] and pseudo-labelled data, with metric variants fine-tuned on Hypersim [15] (indoor) and Virtual KITTI (outdoor). Depth Anything 3 [2] predicts depth and camera rays from any number of views with a plain DINOv2 trans-

former; its monocular metric model predicts depth relative to the focal length. Depth Pro [3] predicts metric depth and focal length from a multi-scale ViT at 1536^2. MoGe-2 [4] predicts an affine-invariant point map, from which it recovers the focal length and shift, plus a global metric scale from the class token. Metric3D v2 [5] predicts depth in a canonical camera space with a fixed focal length and rescales it with the true one; UniDepth V2 [6] predicts a camera representation together with depth and can take the true intrinsics as input. Tile-based methods such as PatchFusion [16] and PatchRefiner [17] target high-resolution monocular metric depth by processing overlapping crops and fusing them with a global estimate.

Stereo depth. SGM [7] remains the workhorse of embedded stereo, with open-source GPU implementations such as libSGM [18]. Learned matchers such as RAFT-Stereo [19] and FoundationStereo [20] are far more accurate but expensive. Fast-FoundationStereo [9] distils FoundationStereo into a real-time model. NVIDIA's ESS [8], distributed with Isaac ROS, is a compact learned stereo network designed for robots on Jetson. Embedded GPUs also ship dedicated stereo hardware: the Orin's optical flow accelerator (OFA) computes SGM-style disparity, exposed through NVIDIA VPI; we do not evaluate it.

Monocular versus stereo. Pseudo-LiDAR [21] showed that image-based depth representation matters for 3D detection, with stereo far ahead of monocular. Depth networks are also known to smooth across depth discontinuities and produce "flying pixels" between foreground and background. We are not aware of a study that compares how the metric accuracy of current monocular foundation models and of stereo changes with input resolution, or that compares them under a common embedded compute budget.

3 Methods compared

Each method runs once per frame on the full-resolution Middlebury images. Methods that cannot process the full 5.7 MP image on the 64 GB Orin receive the images downsampled to the largest input they support. Because the monocular networks are trained at much smaller inputs, we also run each of them at its native input size, and we run libSGM and Fast-FoundationStereo at comparably small inputs so that both sensor types are compared at low compute as well. Every method that can therefore runs at two or three input sizes, which gives the resolution-scaling curves of section 6.1: Depth Anything V2-S and -L, Depth Anything 3, MoGe-2, Metric3D v2 and UniDepth V2 also run at about twice their native size in each dimension, and libSGM at full, half and quarter resolution. ESS has a fixed input size. Table 1 lists the processing resolutions, disparity search range and limiting factor of each method.

libSGM. The open-source CUDA semi-global matcher of Fixstars [18] (version 3.1.0, Apache-2.0). It computes a per-pixel Hamming cost on symmetric 9×7 census codes, aggregates it along four scan paths (horizontal and vertical), selects the winner with a uniqueness test and parabolic sub-pixel refinement, applies a 3×3 median filter and invalidates pixels that fail a left–right consistency check. It stores the aggregated cost volume per path with 64-bit offsets, so it runs at full resolution. We run it on grey-scale input with 256 disparity levels at full resolution and at half and quarter resolution (area downsampling); the 256 levels cover 255 px of full-resolution disparity at full resolution, ≈510 px at half and ≈1020 px at quarter resolution. Its smoothness penalties P_1 and P_2 are tuned separately for each resolution (appendix C); all other parameters keep libSGM's defaults.

Fast-FoundationStereo (FFS). Official research checkpoints `23-36-37` (8 GRU iterations, most accurate) and `20-30-48` (4 iterations, fastest), exported with the authors' single-ONNX script. The largest input that fits on the 64 GB Orin is 768×1152: at 960×1440 TensorRT finds no implementation for the cost-volume pooling, and at 1280×1920 the ONNX export runs out of host memory. To show how FFS scales, we also run both checkpoints at 512×768 and 384×576. The maximum disparity is $0.27\times$ the input width rounded up to a multiple of 32 (320, 224 and 160 px; ≈800 px at full resolution), which covers the dataset. We also benchmark C-Fast-FoundationStereo, the commercially licensed checkpoint (`nvidia/c-fast-foundationstereo`), at the same three sizes: it has the architecture and configuration of `23-36-37` with separately trained weights.

Depth Pro. Fixed 1536×1536 network input. Its metric depth needs a focal length: we evaluate it both with the *true* focal length from the Middlebury calibration and with Depth Pro's own field-of-view estimate.

Depth Anything V2 (metric). The Hypersim (indoor) metric checkpoints, which suit Middlebury's indoor scenes, at the Small, Base and Large sizes, run at full resolution (1988×2884, i.e. 142×206 patches of 14 px) and, for reference, at their native 518×770 input (the 518 px short side they are trained at) and, for Small and Large, at 1008×1512. Exporting at full resolution requires memory-efficient attention (identical computation); the TensorRT engines take 27–79 min to build and use most of the Orin's 64 GB.

Depth Anything 3 Metric-Large. The only monocular metric model of the Depth Anything 3 family (0.35 B

Table 1: Configuration of each method. All methods receive the same full-resolution (≈5.7 MP) images and are scored against full-resolution ground truth; methods that cannot process 5.7 MP on the 64 GB Jetson AGX Orin run at their maximum input size. Full-resolution equivalents assume a 2880 px wide image.

Method	Processing resolution	Full res.?	Max. disparity search range	Limiting factor	Licence (as stated by the authors)
Stereo					
libSGM (Fixstars, 4 paths)	full (≈1990×2880); also half (≈995×1440) and quarter (≈497×720)	yes	256 levels at the processing resolution (256, ≈512 and ≈1024 px at full resolution)	—	Apache-2.0
ESS, Light ESS (NVIDIA Isaac ROS)	576×960 (ESS), 288×480 (Light ESS)	no	no fixed limit (regression)	fixed network input	NVIDIA TAO model EULA (commercial use permitted)
Fast-Foundation-Stereo (both checkpoints)	768×1152; also 512×768 and 384×576	no	0.27× input width: 320, 224 and 160 px (≈800 px at full resolution)	does not fit the 64 GB Orin at larger sizes: no TensorRT implementation for the cost-volume pooling at 960×1440; ONNX export runs out of host memory at 1280×1920	NVIDIA Source Code License (non-commercial)
C-Fast-Foundation-Stereo (commercial checkpoint)	768×1152; also 512×768 and 384×576	no	as Fast-FoundationStereo	as Fast-FoundationStereo	NVIDIA Open Model Agreement
Monocular					
Depth Pro	1536×1536	no	n/a	fixed network input	Apple licence
Depth Anything V2 (S, B, L)	full (1988×2884); also native 518×770 and (S, L) 1008×1512	yes	n/a (metric head limited to 20 m)	— (trained at 518 px; FP16 engine of the Small model is numerically unreliable at full size, section 6.5)	S: Apache-2.0; B, L: CC BY-NC 4.0
Depth Anything 3 Metric-L	native 350×504 (longest side 504 px); also 700×1008 and full (1988×2884)	yes	n/a	— (trained at 504 px)	Apache-2.0
MoGe-2 (S, B, L)	native ≈700×1008 (3600 tokens, default level); also 1400×2030 and full (1988×2884)	yes	n/a	— (token count is an input parameter)	MIT
Metric3D v2 (ViT-S, ViT-L)	native 616×1064; also 1232×2128 and full (1988×2884)	yes	n/a	— (input size is a preprocessing parameter)	BSD-2-Clause
UniDepth V2 (S, B, L)	native 644×938 (default pixel budget); also 1302×1890 and full (1988×2884)	yes	n/a	— (pixel budget is a parameter)	CC BY-NC 4.0

parameters, ViT-L, Apache-2.0). Its official preprocessing resizes the longest side to 504 px and rounds both sides to multiples of 14 (350 × 504, or 336 × 504 for the scenes with a wider aspect ratio); we also set the longest side to 1008 px (≈700 × 1008) and run each image at its own full size, rounded to multiples of 14. Its output becomes metric depth by multiplying with the focal length in pixels of the processed image and dividing by 300; we use the true focal length.

MoGe-2. The Small, Base and Large checkpoints (ViT-S/B/L, the "-normal" variants) at their default res-

olution level, 3600 patch tokens ($\approx 700 \times 1008$ for Middlebury's 3:2 images), and at 14,400 tokens ($\approx 1400 \times 2030$) and full resolution (one token per 14×14 px, ≈29,000 tokens), with the predicted metric scale. Like Depth Pro we evaluate MoGe-2 both with the true focal length, from which it recovers only the depth shift, and with its own focal-length estimate. We do not apply its validity mask, so its output is dense like that of the other monocular models.

Metric3D v2. The ViT-Small and ViT-Large models with the official preprocessing: the image is resized to fit the 616×1064 input keeping its aspect ratio and padded, and the depth predicted in a canonical camera with a 1000 px focal length is rescaled with the true focal length of the resized image. We also use inputs of 1232×2128 and 1988×2884 (full resolution). Its ViT attention is computed in blocks of queries at full resolution, which is the same computation in less memory.

UniDepth V2. The Small, Base and Large models through the official `infer`, given the true intrinsics. UniDepth chooses its processing size from a pixel budget (0.2–0.6 MP by default, 644×938 for Middlebury); we also set budgets that give 1302×1890 and full resolution. Its TensorRT engines take the camera as rays (the official `UniDepthV2ONNXcam` export).

ESS and Light ESS. NVIDIA's stereo networks from Isaac ROS (NGC model version 4.1.0), with fixed inputs of 576×960 (ESS) and 288×480 (Light ESS); custom layers come from the TensorRT plugin library distributed with the model. Both images are resized to the network input, which changes their aspect ratio from 3:2 to 5:3 but not the horizontal disparity, and normalised as on the model card. We use the dense disparity output, without the optional confidence threshold.

4 Experimental setup

Platform. NVIDIA Jetson AGX Orin 64GB developer kit, JetPack 6.2 (L4T R36.4.3), CUDA 12.6, TensorRT 10.3, MAXN power mode. All networks are timed as TensorRT FP16 engines built on the device. Monocular accuracy comes from the FP32 PyTorch models with the official pre- and post-processing, except for Depth Anything V2 at native input and V2-B/L at full resolution, whose FP16 engines (and those at 1008×1512) agree with FP32 to within 0.25% and are used for both (section 6.5). ESS and Light ESS are evaluated with their TensorRT engines, the only form in which they are distributed.

Data. Middlebury 2014 [10] with "perfect" rectification at full resolution ($\approx 2.9\text{k} \times 2.0\text{k}$). Of the 23 scenes with public ground truth we use the 10 whose non-occluded ground-truth disparity stays below 256 px (at most 243 px), so that the 256-level semi-global matcher is not limited by its disparity range at any of the resolutions we test; appendix A lists them. In the other 13 scenes near objects reach 265–712 px of disparity at full resolution.

Non-occluded ground truth. Middlebury's ground truth comes from structured light, not from stereo matching, so it also covers pixels that the right camera cannot see: on average 5.8% of the ground-truth pixels are occluded in the right view and 2.7% fall outside its field of view. No stereo method can match these pixels, so, following MiddEval3, we evaluate every method only on the *non-occluded* ground truth (91% of it): pixels whose match lies inside the right image and whose right-view ground-truth disparity agrees within 1 px. Monocular methods are evaluated on the same pixels.

Protocol. Downsampled outputs are upsampled to full resolution, bilinearly for the dense network outputs and with nearest-neighbour interpolation for libSGM, so that its holes stay holes; disparities are rescaled by the width ratio. Depth follows from disparity via $Z = fB/(d + d_{\text{offs}})$; monocular depth is converted to disparity the same way, so a monocular pixel predicted beyond the zero-disparity distance fB/d_{offs} counts as missing.

Metrics. For disparity: bad-t, the percentage of non-occluded ground-truth pixels with error $> t$ px, where missing predictions count as errors. For depth [22]: AbsRel and the threshold accuracy $\delta < 1.05$ (fraction of pixels within 5% of the true depth), over the predicted pixels. The uniqueness and left–right checks of libSGM leave holes, so we report it both raw (density, plus errors on predicted pixels) and after MiddEval3-style background hole filling (each hole takes the smaller of the nearest valid disparities to its left and right on the same row). For monocular methods we additionally report *oracle-scaled* metrics, where each prediction is multiplied by the per-image median ratio to the non-occluded ground truth, an upper bound on what any scale-recovery scheme could achieve.

Accuracy, not precision. All metrics above measure *accuracy*: the error of each depth value with respect to the ground truth. We deliberately do not report *precision*, i.e. the noise or repeatability of the depth map around its own local or temporal mean, because metric accuracy is what downstream tasks such as obstacle detection, ground-plane estimation and free-space computation depend on. Precision is particularly misleading for neural networks. A network can produce a perfectly smooth, noise-free surface that is nevertheless in the wrong place: in our experience with field data, a monocular network renders a flat floor as a perfectly flat plane with no visible noise, yet with its tilt off by several degrees. A precision

metric would rate such a hallucinated surface as excellent; an accuracy metric against independent ground truth exposes the error. The oracle-scaled metrics remove only a single global scale factor per image, so such structural errors (tilt, warping, misplaced objects) remain in them as well. Classical stereo shows the opposite pattern: noisier depth values and missing pixels where matching fails, but on the pixels it does return, its error is governed by matching and triangulation geometry rather than by a learned prior of what the scene should look like.

Speed and energy. For networks we report the median `trtexec` GPU compute time (spin-wait, 3 s warm-up, ≥20 s run). For libSGM we report the median per-frame GPU busy time from an Nsight Systems trace: the union of all kernel and copy intervals, including the upload of both images and the download of the disparity. Power is sampled with `tegrastats` (VDD_GPU_SOC + VDD_CPU_CV); dynamic energy per frame is $(P_{\text{load}} - P_{\text{idle}})$/throughput.

5 Which method to use

Table 2 summarises the results as recommendations for a single Jetson AGX Orin.

With two cameras. At 30 fps, ESS is the best choice: 0.70% AbsRel with a dense output in 8.6 ms, and Light ESS 1.16% in 2.8 ms, which leaves room for several camera pairs or other networks. libSGM at half resolution is about as accurate (0.73% in 25 ms) on the 85% of the pixels it returns, is open source under Apache-2.0 and can be tuned or modified. Where GPU time is the scarcest resource, libSGM at quarter resolution is the open-source choice: 1.18% AbsRel on 88% of the pixels in only 6.1 ms and 0.18 J per frame, so that at 30 fps it occupies 18% of the Orin's GPU and leaves the rest to other tasks; only Light ESS needs less (2.8 ms for 1.16%, dense). At 10 fps, Fast-FoundationStereo with 4 iterations at 384 × 576 reaches 0.64% in 68 ms, and with more compute 0.33% (8 iterations, 768 × 1152, 758 ms); its research checkpoints are licensed for non-commercial use only. The commercially licensed C-Fast-FoundationStereo is as accurate as the research checkpoint of the same architecture (0.33% at 768 × 1152 in 760 ms, 0.52% at 384 × 576 in 116 ms), so for a commercial product it is the choice when more than 100 ms per frame is available.

With one camera. No monocular model comes close to stereo in metric accuracy. The most accurate are Depth Anything 3 Metric-Large at its native input (12.3% in 33 ms, Apache-2.0) and Depth Pro (12.0% in 803 ms), both given the true focal length. If the metric scale can be obtained from another source, such as a known object size, a range sensor or odometry, the best shapes come from MoGe-2-L (3.8% with oracle scale, 212 ms, MIT), UniDepth V2-B (3.9%, 85 ms, non-commercial) and Metric3D v2-L (4.0%, 265 ms, BSD-2-Clause), and at 30 fps from Depth Anything 3 (6.0%). Running the monocular models above their native input does not help (section 6.1).

6 Results

6.1 Resolution scaling

Figure 1 plots AbsRel against the number of pixels each method processes, for every method that runs at more than one input size. Stereo error falls with every increase in resolution. libSGM goes from 1.18% at quarter resolution (0.36 MP) to 0.73% at half and 0.63% at full resolution (5.7 MP); Fast-FoundationStereo from 0.51% to 0.42% and 0.33% with 8 iterations and from 0.64% to 0.51% and 0.40% with 4, between 0.22 and 0.88 MP. The gains are smaller than the ideal of the triangulation argument, which halves the error with every doubling of the image side at a constant matching error in pixels; at the last step libSGM gains 14%. The matching error in pixels grows at higher resolution, most likely because a fixed-size matching window then covers less of the scene.

No monocular model improves with resolution. Depth Anything 3 goes from 12.3% at its native 350 × 504 input to 14.2% at 700 × 1008 and 54.9% at full resolution, MoGe-2-L from 19.2% to 23.4% and 24.4%, Depth Anything V2-L from 32.5% to 39.5% and 42.7%, and V2-S from 29.9% to 43.8% and 56.0%. Metric3D v2 collapses above its native input: its error rises from 57–60% to 128–162%, and it places 86–95% of the pixels beyond the zero-disparity distance. UniDepth V2 stays at 20–28% at every size, slightly better at full resolution for the Small model (24.7% to 22.6%) and worse for the Large model (20.5% to 28.1%). The cause is the metric scale (table 3). Depth Anything V2 places the scenes 1.45–1.54 times too close at its native input and 1.8–2.4 times too close at full resolution, and Depth Anything 3 is almost unbiased at its native input (scale 0.98) but places the scenes 1.5 times too far at full resolution. The models are trained at 0.2–0.7 MP; at full resolution each 14 px patch covers much less of the scene, the token count is 8–32 times larger and the positional embeddings are interpolated far beyond the training range, and nothing in the networks ties the metric scale to a quantity that becomes more precise with more pixels.

With the scale removed by an oracle (fig. 1b) the shape does not improve either: the oracle-scaled AbsRel stays flat or rises, from 6.0% to 6.2% for Depth Anything 3, from 3.8% to 4.0% for MoGe-2-L, from 8.3–8.9% to 9.7–10.7% for Depth Anything V2, from 3.9–5.7% to 6.5–9.0% for UniDepth V2 and from 4.0–6.5% to 5.2–8.4% for Metric3D v2. Higher resolution sharpens the monocular depth maps (section 6.4), but their error is dominated by smooth deviations in scale, tilt and curvature that come from the network's global interpretation of the scene, not

Table 2: Recommendations: the most accurate configuration whose GPU time on one Jetson AGX Orin (MAXN, FP16) fits the frame-rate budget, by AbsRel on the non-occluded pixels of the 10 Middlebury 2014 scenes. Monocular models with the true focal length or intrinsics; the last column assumes the metric scale is supplied by an oracle (an upper bound). Last row: the configuration with the lowest GPU time among the methods whose licences permit commercial use and modification (libSGM, Depth Anything V2-S and 3, MoGe-2, Metric3D v2). Licences in table 1.

Budget	Best stereo AbsRel (GPU time)	Best monocular AbsRel (GPU time)	Best monocular, oracle scale AbsRel (GPU time)
30 fps	ESS (NVIDIA), 576×960: 0.70% (9 ms)	Depth Anything 3 Metric-L, 350×504: 12.34% (33 ms)	Depth Anything 3 Metric-L, 350×504: 5.97% (33 ms)
10 fps	Fast-FoundationStereo 20-30-48, 384×576: 0.64% (68 ms)	Depth Anything 3 Metric-L, 350×504: 12.34% (33 ms)	UniDepth V2-B, 644×938: 3.90% (85 ms)
1 fps	C-Fast-FoundationStereo, 768×1152: 0.33% (760 ms)	Depth Pro, 1536×1536: 11.98% (803 ms)	MoGe-2-L, 700×1008: 3.78% (212 ms)
no limit	C-Fast-FoundationStereo, 768×1152: 0.33% (760 ms)	Depth Pro, 1536×1536: 11.98% (803 ms)	MoGe-2-L, 700×1008: 3.78% (212 ms)
lowest GPU, open source	libSGM, ≈497×720: 1.18% (6 ms)	Depth Anything V2-S (native), 518×770: 29.88% (12 ms)	Depth Anything V2-S (native), 518×770: 8.93% (12 ms)

Table 3: Monocular models across input sizes. Scale: median over the scenes of the per-image median ratio of ground-truth to predicted depth (1 = unbiased, >1 = predicted too close). Density below 100% means that part of the scene is placed beyond the zero-disparity distance.

Model	Input	MP	Scale GT/pred.	AbsRel [%]	Oracle AbsRel [%]	Density [%]
Depth Anything V2-S	518×770	0.40	1.45	29.9	8.9	100
Depth Anything V2-S	1008×1512	1.52	1.88	43.8	9.2	100
Depth Anything V2-S	1988×2884	5.73	2.44	56.0	9.7	100
Depth Anything V2-B	518×770	0.40	1.50	30.6	8.9	100
Depth Anything V2-B	1988×2884	5.73	1.83	41.2	10.6	100
Depth Anything V2-L	518×770	0.40	1.54	32.5	8.3	100
Depth Anything V2-L	1008×1512	1.52	1.78	39.5	9.9	100
Depth Anything V2-L	1988×2884	5.73	1.91	42.7	10.7	100
Depth Anything 3 Metric-L	350×504	0.18	0.98	12.3	6.0	93
Depth Anything 3 Metric-L	700×1008	0.71	0.93	14.2	6.1	90
Depth Anything 3 Metric-L	1988×2884	5.73	0.66	54.9	6.2	56
MoGe-2-S	700×1008	0.71	0.78	33.3	4.9	76
MoGe-2-S	1400×2030	2.84	0.73	42.9	5.6	68
MoGe-2-S	1988×2884	5.73	0.76	40.2	6.0	66
MoGe-2-B	700×1008	0.71	0.83	21.6	3.9	96
MoGe-2-B	1400×2030	2.84	0.76	34.4	4.4	86
MoGe-2-B	1988×2884	5.73	0.74	37.2	4.5	81
MoGe-2-L	700×1008	0.71	0.89	19.2	3.8	94
MoGe-2-L	1400×2030	2.84	0.85	23.4	4.1	87
MoGe-2-L	1988×2884	5.73	0.85	24.4	4.0	86
Metric3D v2-S	616×1064	0.66	0.70	60.1	6.5	65
Metric3D v2-S	1232×2128	2.62	0.47	133.0	7.9	14
Metric3D v2-S	1988×2884	5.73	0.40	162.5	8.4	5
Metric3D v2-L	616×1064	0.66	0.65	56.9	4.0	63
Metric3D v2-L	1232×2128	2.62	0.47	127.5	5.6	9
Metric3D v2-L	1988×2884	5.73	0.45	136.8	5.2	8
UniDepth V2-S	644×938	0.60	0.97	24.7	5.7	93
UniDepth V2-S	1302×1890	2.46	1.06	23.4	8.0	95
UniDepth V2-S	1988×2884	5.73	1.15	22.6	9.0	96
UniDepth V2-B	644×938	0.60	0.91	26.0	3.9	84
UniDepth V2-B	1302×1890	2.46	1.15	21.2	5.8	95
UniDepth V2-B	1988×2884	5.73	1.37	26.2	7.7	96
UniDepth V2-L	644×938	0.60	1.15	20.5	4.1	99
UniDepth V2-L	1302×1890	2.46	1.28	25.0	5.6	100
UniDepth V2-L	1988×2884	5.73	1.34	28.1	6.5	100

from missing image detail.

6.2 Accuracy

Table 7 in appendix B lists accuracy, GPU time and energy for every configuration, averaged over the 10 scenes.

Stereo accuracy. Fast-FoundationStereo is the most accurate method: 0.33% AbsRel with 8 iterations and 0.40% with 4 at 768×1152, at 100% density and 99% of pixels within 5% of the true depth. It degrades gracefully with input size: each step down to 512×768 and 384×576 cuts its GPU time by 2–3 times and adds only about 0.1 percentage points of AbsRel (0.51% and 0.64% at 384×576). libSGM reaches 0.63% AbsRel on 76% of the non-occluded pixels at full resolution, 0.73% on 85% at half and 1.18% on 88% at quarter resolution; each halving of the image side cuts its GPU time by about four. At full resolution libSGM is about as accurate as Fast-FoundationStereo at 384×576 (0.64%) on the pixels it returns, at 1.5 times the GPU time but without its density. After hole filling, libSGM reaches 1.92%, 1.79% and 2.13% at full, half and quarter resolution. The tuned smoothness penalties (appendix C) lower libSGM's AbsRel by 39%, 30% and 9% relative to its defaults, at 4–8 percentage points lower density. NVIDIA's ESS is dense and reaches 0.70% AbsRel at its fixed 576×960 input, and Light ESS 1.16% at 288×480, the best accuracy per unit of compute of all methods. Their bad-2 rates (20% and 37%) are higher than those of Fast-FoundationStereo (7–17%) because a disparity error of one pixel at their input is three or six pixels at full resolution. The commercially licensed C-Fast-FoundationStereo matches the research checkpoint `23-36-37`, whose architecture it shares, to within 0.01 percentage points of AbsRel at every input size (0.33%, 0.43% and 0.52%), with a slightly lower bad-2 rate at 768×1152 (6.0% against 6.5%) and the same GPU time.

Monocular accuracy. No monocular model is close to metric accuracy at Middlebury's 1.3–7.6 m depth range. The best raw results come from Depth Pro with the true focal length, 12.0% AbsRel (19.0% with its own focal-length estimate), and from Depth Anything 3 Metric-Large, 12.3% at a fraction of the compute; only 29% and 31% of their pixels are within 5% of the true depth. MoGe-2 reaches 19–33% with the true focal length and 30–40% with its own: its metric scale is unreliable on these scenes, and with the true focal length it places parts of some scenes beyond the zero-disparity distance (76–97% density). Depth Anything V2 has 30–32% AbsRel at its native input and 41–56% at full resolution, where only 0.1–2% of its pixels are within 5% of the true depth. None of the monocular models benefits from a larger input: Depth Anything 3 has 14.2% AbsRel at 700×1008 and 54.9% at full resolution, where it places 44% of the pixels beyond the zero-disparity distance, and MoGe-2 with the true focal length has 23–43% at 1400×2030 and 24–40% at full resolution, while its oracle-scaled error rises by 0.2–1.1 percentage points. All numbers below are at native input unless stated otherwise. Removing the global scale error with an oracle helps all of them, most of all MoGe-2, whose point maps have the best shape: with

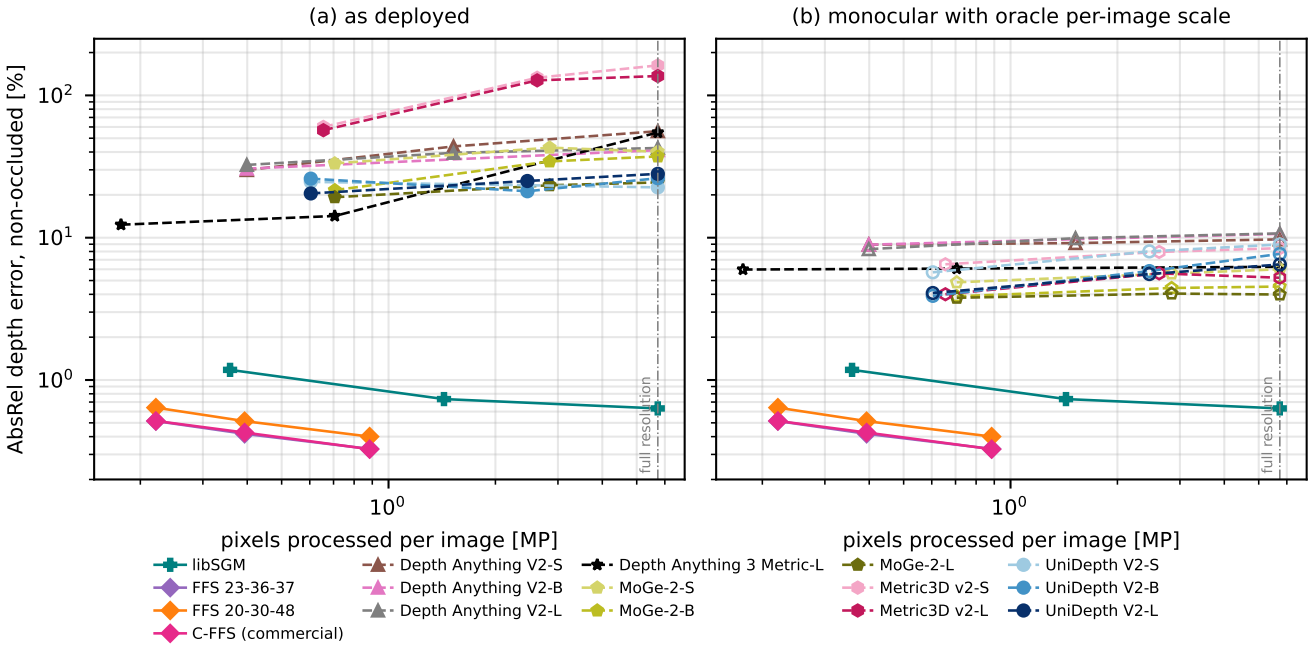


Figure 1: Resolution scaling. AbsRel (non-occluded pixels, mean over the 10 scenes, log scale) against the number of pixels each method processes, for every method run at more than one input size. (a) As deployed. (b) Monocular models after per-image oracle median scaling (hollow markers), stereo unchanged. Stereo (solid lines): libSGM at quarter, half and full resolution and Fast-FoundationStereo (FFS) at 384×576, 512×768 and 768×1152, its largest input on the Orin (ESS has a fixed input). Monocular (dashed lines): each model from its native input to full resolution, with the true focal length or intrinsics. Stereo error falls with resolution; monocular error does not, and with an oracle scale it stays flat or rises.

oracle scale MoGe-2-L reaches 3.8%, Depth Pro 4.7%, Depth Anything 3 6.0% and Depth Anything V2 8.3–8.9%. Even the best of these keeps 27% of its pixels more than 5% from the true depth. Of the two further families, UniDepth V2, given the true intrinsics, has 20–26% AbsRel at its native input (3.9–5.7% with oracle scale), and Metric3D v2, given the true focal length, 57–60% (4.0–6.5%): it places most scenes too far and 35–37% of the pixels beyond the zero-disparity distance.

Stereo versus monocular. Every stereo method is more accurate than every monocular method, raw or hole-filled, even when the monocular models are given an oracle scale: the worst stereo result (hole-filled libSGM at quarter resolution, 2.13%) is below the best oracle-scaled monocular result (MoGe-2-L, 3.78%). Without the oracle, the most accurate monocular model per unit of compute, Depth Anything 3, has 10 times the AbsRel of libSGM at quarter resolution, which needs a fifth of its GPU time, and 18 times that of ESS, which needs a quarter.

6.3 Depth discontinuities

AbsRel averages the error over all non-occluded pixels and is therefore dominated by large surfaces. Depth discontinuities, where obstacle detection, grasping and free-space estimation need clean object boundaries, make up only 1.7% of the scored pixels. We analyse them separately (table 4, fig. 2). Each method is evaluated at its own output resolution, upsampled with nearest-neighbour interpolation so that no intermediate depths come from our resampling, and monocular depth with its oracle scale, so that a global scale error is not counted as an edge error.

Table 4: Depth discontinuities, mean over the 10 scenes. Edge pixels: non-occluded ground-truth pixels whose 17×17 neighbourhood spans a depth ratio of at least 1.2 (1.7% of the scored pixels). Flying points: predicted edge pixels more than 5% in front of the farthest and behind the nearest ground-truth surface in that neighbourhood. Each method at its own output resolution (nearest-neighbour upsampling); monocular depth with oracle scale.

Method (output resolution)	Density at edges [%]	Flying points [%]	AbsRel at edges [%]	AbsRel elsewhere [%]
libSGM, full res.	45	1.1	4.33	0.59
libSGM, quarter res.	59	9.4	9.08	1.08
FFS 23-36-37, 768×1152	100	3.0	2.48	0.29
FFS 20-30-48, 768×1152	100	4.1	2.85	0.36
C-FFS, 768×1152	100	2.4	2.42	0.29
Depth Anything V2-S, 518×770	100	25.8	11.43	8.90
Depth Anything V2-L, 518×770	100	20.3	11.94	8.26
Depth Anything 3 Metric-L, 350×504	100	16.6	10.25	5.92
MoGe-2-L, 700×1008	100	46.1	7.46	3.71
ESS, 576×960	100	3.8	4.51	0.63
Light ESS, 288×480	100	8.8	7.60	1.03
Metric3D v2-L, 616×1064	100	22.8	8.53	3.93
UniDepth V2-L, 644×938	100	17.1	7.83	4.03

The monocular models smooth across discontinuities

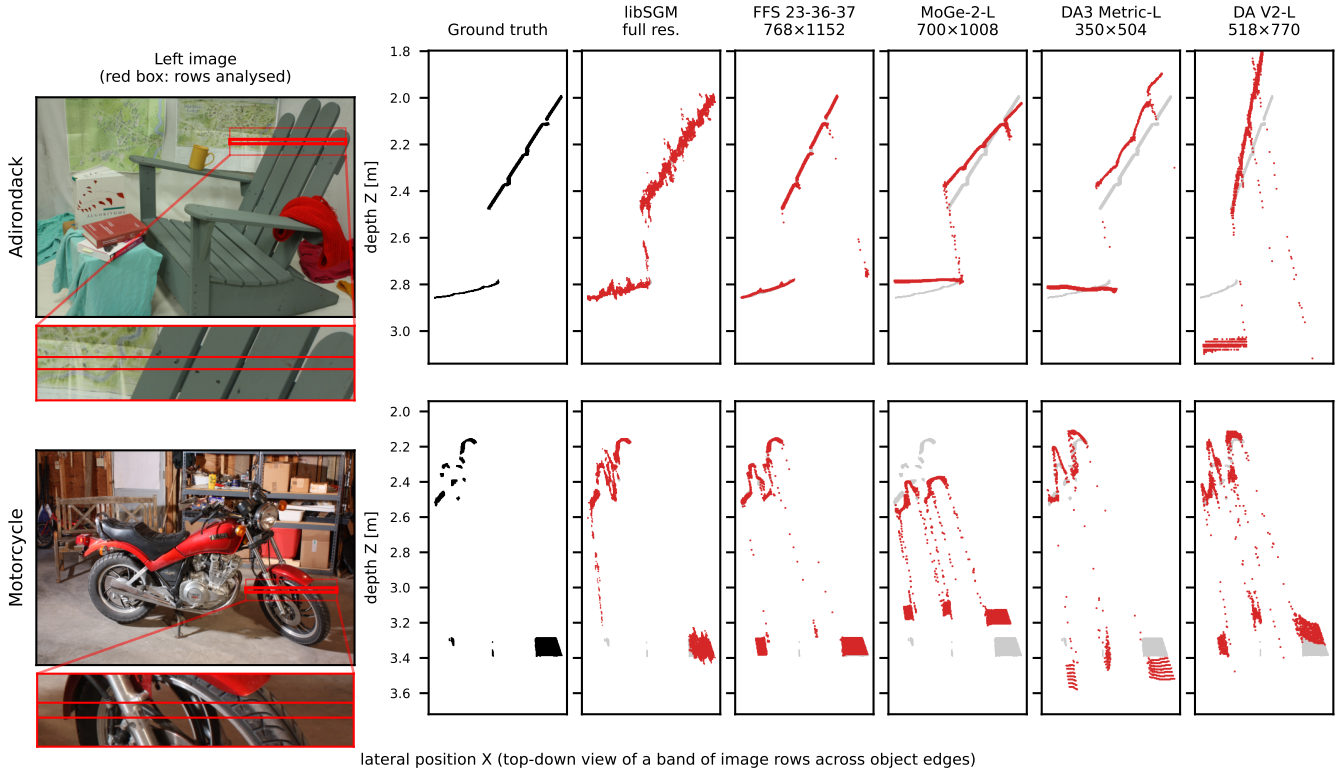


Figure 2: Point clouds at object edges, seen from above: depth Z against lateral position X of the points in a band of 40 image rows across the back slats of the Adirondack chair (top) and the front wheel of the motorcycle (bottom); the left column shows the rectified left image with the band in red and, below it, a magnified view of the band with some rows of context. Grey: ground truth. Each method is shown at its own output resolution (one point per output pixel); monocular depth with oracle scale. The monocular models connect foreground and background with ramps of "flying" points; Fast-FoundationStereo keeps dense, clean surfaces with a few flying points; libSGM at full resolution has the fewest flying points but noisier surfaces and missing points at the edges.

and connect foreground and background with continuous ramps of points (fig. 2): 46% of MoGe-2-L's edge points are flying points, 20–26% of Depth Anything V2's, 23% of Metric3D v2-L's and 17% of Depth Anything 3's and UniDepth V2-L's. Fast-FoundationStereo returns a point for every pixel and keeps clean surfaces, with 2.4–4.1% flying points (2.4% for C-Fast-FoundationStereo), mostly at thin structures such as the spokes of the wheel. libSGM at full resolution has the fewest flying points of all methods, 1.1%: it does not interpolate across an edge but rejects ambiguous pixels in its left–right check, so it returns only 45% of the edge pixels, and its surfaces are noisier. At quarter resolution libSGM has 9.4% flying points, more than Fast-FoundationStereo. ESS has 3.8%, similar to Fast-FoundationStereo, and Light ESS, at a quarter of ESS's input pixels, 8.8%, similar to libSGM at quarter resolution.

AbsRel registers almost none of this. MoGe-2-L's AbsRel is 7.5% at edge pixels and 3.7% elsewhere, but with 1.7% of the pixels at edges, its flying points add less than 0.1 percentage points to its overall error, and the oracle-scaled error that ranks MoGe-2 as the best monocular shape is the same error that hides its edge behaviour. A dense, accurate-on-average depth map can therefore still misplace object boundaries, and a classical matcher can be the better choice where boundaries matter: at full resolution, libSGM's discontinuities are cleaner than those of the learned stereo network, at the cost of missing pixels, a trade-off that a mean error over predicted pixels does not show.

6.4 Qualitative results

Figure 3 shows depth and error maps on two scenes for every method at full resolution, or at its maximum or fixed input size, with the monocular networks after oracle scaling, i.e. with their global scale error removed. Stereo errors are mostly below 1% and concentrate at depth discontinuities and thin structures (leaves, straw); libSGM leaves holes there and on low-texture walls, and its full-resolution output is speckled where the left–right check rejects isolated pixels. The oracle-scaled monocular

depth maps look plausible and correctly ordered, yet keep 2.7–9.3% error that is spatially structured: smooth bands across the walls and floor of Classroom1 that correspond to tilted or warped surfaces. This is the failure mode that a precision metric would not see (section 4). Without the oracle scale (table 7), the monocular errors are far larger.

Figure 4 compares the methods near the monocular models' native input, as deployed, i.e. without oracle scale. libSGM at quarter resolution (6.1 ms) keeps about 1% error with holes at occlusions and in low-texture regions; Fast-FoundationStereo at 512×768 is dense and has 0.35–0.48% error. The monocular models at their native input have 11–39% error on these scenes: Depth Anything V2 places almost all of Sword1 at the near end of the depth range and MoGe-2-L at the far end, and the error of Depth Anything 3 falls to 2.0–9.9% only with the oracle scale.

6.5 FP16 fidelity

We compared every monocular TensorRT FP16 engine with its FP32 PyTorch model on one scene. Depth Anything V2 at native resolution differs by 0.10–0.17%, and Base and Large at full resolution by 0.19% and 0.25% (median relative depth difference). Depth Anything V2-Small at full resolution differs by 17.5%, and Depth Pro by 1.9% (with a 1.6° shift of its field-of-view estimate). For these two we report accuracy from FP32 and speed from FP16, which favours them. The FP32 Small model is itself inaccurate at this resolution (57.7% AbsRel), so the FP16 error is not the cause of its poor result.

For Depth Anything 3 and MoGe-2 we found two TensorRT 10.3 problems, both in fused kernels: with every intermediate tensor marked as an output the engines are correct, and ONNX Runtime reproduces PyTorch to within 0.002%. First, the attention pattern that PyTorch exports from `scaled_dot_product_attention` (queries and keys each scaled by the square root of the scale factor) produced a wrong engine; exporting the same computation as $\mathrm{softmax}(q\,s\,k^{\top})\,v$ fixed it. The Depth Anything V2 full-resolution engines use the same attention export; Base and Large match FP32, but the 17.5% deviation of the Small engine is consistent with this problem, so its timing may not represent a correct engine. Second, MoGe-2-S and -B's three-layer metric-scale head on the class token came out wrong even in FP32; we run this negligible computation outside the engine. After these fixes the Depth Anything 3 engine reproduces the FP32 AbsRel to within 0.2 percentage points on the check scenes processed at its 350×504 input size, and the MoGe-2 engines differ from FP32 by 0.03–1.1% (median relative depth). For the larger inputs, up to ≈29,000 tokens at full resolution, the export processes attention in blocks of 2,048 queries, which is the same computation without tracing the full attention matrix. Their accuracy is reported from FP32 and their speed from the fixed FP16 engines.

The UniDepth V2 engines, exported with the same attention form, agree with FP32 to within 0.05–0.49% (median relative depth difference). The Metric3D v2 engines differ by 1.3–3.3% (99th percentile up to 15%), probably because its iterative refinement decoder accumulates FP16 rounding; for both we report accuracy from FP32 and speed from FP16.

7 Accuracy per compute

Figure 5 plots depth error against GPU time for every configuration (raw outputs only).

Speed. GPU time spans more than three orders of magnitude, from 2.8 ms to 5.3 s per frame. Among the stereo methods, Light ESS takes 2.8 ms and ESS 8.6 ms; libSGM 6.1 ms at quarter, 25 ms at half and 101 ms at full resolution; and Fast-FoundationStereo 68 ms (4 iterations) and 116 ms (8 iterations) at 384×576, rising to 445 ms and 758 ms at 768×1152 (C-Fast-FoundationStereo takes the same time as the 8-iteration research checkpoint). Among the monocular models at their native input, Depth Anything V2 takes 11.5 ms (Small), 26.8 ms (Base) and 84.5 ms (Large), Depth Anything 3 33 ms, UniDepth V2 42 ms (Small), 85 ms (Base) and 182 ms (Large), Metric3D v2 54 ms (Small) and 265 ms (Large) and MoGe-2 73 ms, 103 ms and 212 ms. At twice that size, Depth Anything 3 takes 181 ms, UniDepth V2 0.28–1.25 s, Metric3D v2 0.30–1.56 s and MoGe-2 0.42–1.50 s. At full or fixed resolution, Depth Pro takes 803 ms and the other monocular models 0.76–5.3 s. No monocular model is faster than Light ESS, ESS or libSGM at quarter resolution: the fastest, Depth Anything V2-S at native input, needs 4.1 times the GPU time of Light ESS, Depth Anything 3 12 times, and the monocular models at full or fixed resolution 272–1,890 times.

Energy. Dynamic energy per frame is 0.08 J for Light ESS, 0.31 J for ESS, 0.18, 0.73 and 3.0 J for libSGM at quarter, half and full resolution, and 1.9–22.4 J for Fast-FoundationStereo. The monocular models need 0.5–12 J at their native input and 32–235 J at full or fixed resolution. The networks draw 34–57 W on the GPU+SoC and CPU rails under load.

7.1 Cost of compute versus a second camera

Running a method at F frames/s uses a fraction $u = t_{\mathrm{GPU}} F/1000$ of one Orin, where t_{GPU} is the per-frame GPU time in milliseconds. We price this at the module's volume price P_{SoC} ($2,999 for the AGX Orin 64GB at 1k units [23]). A stereo system with GPU time t_s is cheaper than a monocular system with GPU time t_m whenever

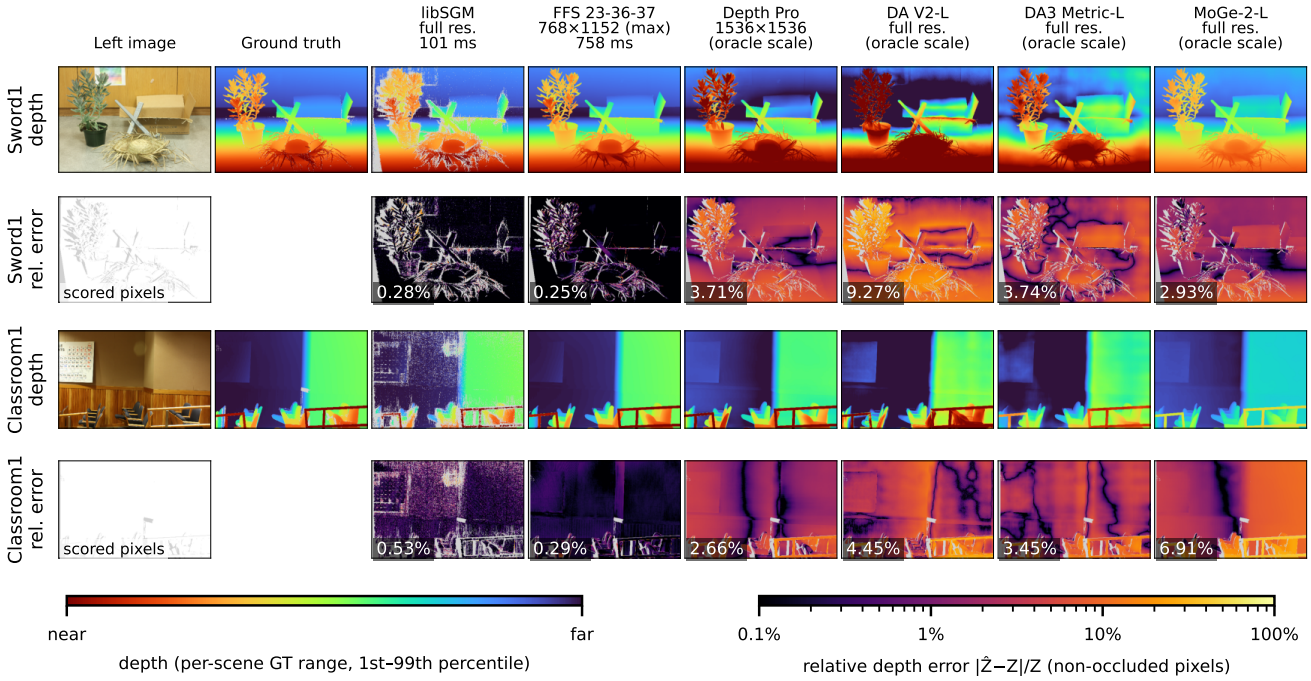


Figure 3: Depth maps (rows 1 and 3) and relative depth error on non-occluded pixels (rows 2 and 4) for Sword1 and Classroom1, with every method at full resolution or at its maximum or fixed input size. Stereo: raw outputs. FFS: Fast-FoundationStereo; DA: Depth Anything. Monocular: depth after per-image oracle median scaling, so the remaining error is structural, not a global scale error. Sword1 is the scene on which every method's error is closest to its median over the 10 scenes; Classroom1 is added for its large planar surfaces. Numbers are the AbsRel of each panel. Depth uses one colour scale per scene (1st–99th percentile of the ground truth); values outside it are clipped. Grey: no ground truth, occluded, or no prediction (stereo holes are not filled). The first column of the error rows shows the scored (non-occluded) pixels in white.

the extra camera costs less than

$$P^*_{\text{cam}} = (t_m - t_s)\,\frac{F}{1000}\,P_{\text{SoC}}. \tag{1}$$

We compare P^*_{cam} with camera prices of \$20 to \$500 per camera; for reference, automotive camera modules average roughly \$20 each across the market [24]. Table 5 applies the model at 10 and 30 fps.

The cheapest stereo configuration, Light ESS, needs 0.08 Orins at 30 fps, ESS 0.26 and libSGM 0.18–3.0; Fast-FoundationStereo needs 2.0–22.7. The monocular models need 0.35 (Depth Anything V2-S) to 8.0 (Metric3D v2-L) Orins at their native input and 22.8 to 159 at full or fixed resolution.

Relative to Light ESS, the break-even camera price at 30 fps is \$786 (Depth Anything V2-S) to \$23,618 (Metric3D v2-L) for the monocular models at their native input and \$68,266 to \$476,366 at full or fixed resolution (\$71,949 for Depth Pro). Every one of them is above the whole \$20–\$500 range of camera prices, so at 30 fps a second camera plus Light ESS is the cheaper system even with a \$500 camera. At 10 fps the thresholds are a third as large; they remain above \$500 for every monocular model except Depth Anything V2-S at native input, whose threshold of \$262 falls within the range. Against ESS, the full- or fixed-resolution thresholds at 30 fps are \$67,748 to \$475,845. Fast-FoundationStereo, the most accurate method, needs more GPU time than several monocular models at native input but has 19–182 times lower AbsRel: learned stereo buys accuracy with compute, while an efficient stereo matcher is what makes the second camera economical.

8 Discussion and limitations

Why stereo scales and monocular depth does not. Stereo measures depth: the disparity of a point is observed in the images, and the calibrated focal length and baseline turn it into metric depth, with an error that shrinks as the pixels become smaller. A monocular network infers depth from learned regularities of the scene, and its metric scale comes from a prior over object sizes and camera geometry seen in training. More pixels add detail but no information about that scale, and outside the training resolution the prior itself shifts (table 3). The two approaches therefore move apart as cameras gain resolution.

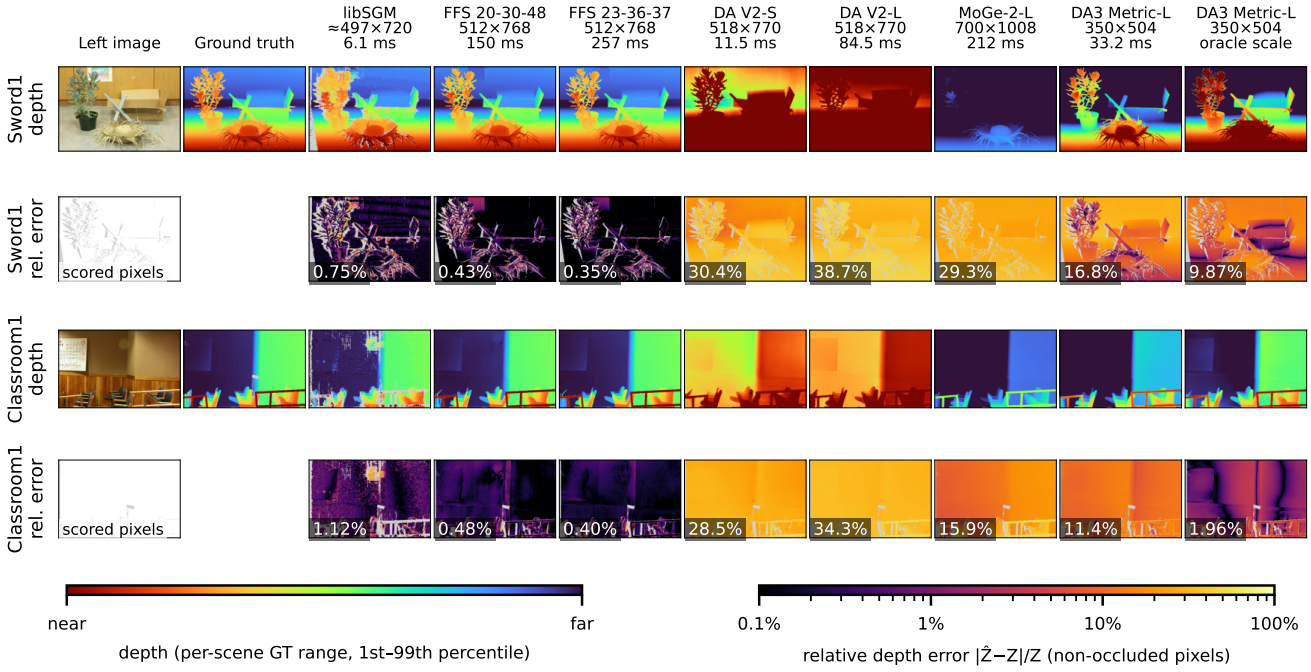


Figure 4: As fig. 3, at low resolution: libSGM at quarter resolution, Fast-FoundationStereo at 512 × 768 and the monocular models at their native input, each as deployed, i.e. monocular depth with its own metric scale (last column: Depth Anything 3 with oracle scale for comparison). Column titles give the processing resolution and GPU time per frame on Jetson AGX Orin.

Running monocular models beyond their training resolution. One may object that the monocular models were never meant to run at 5.7 MP. That is the point of the comparison: stereo turns additional pixels into metric accuracy, and the released monocular models cannot. They are also far less accurate at their native input, where the best of them, Depth Anything 3, has 18 times the error of ESS, which needs a quarter of its GPU time, and still 8.5 times that error with an oracle scale. Depth Pro, whose multi-scale design targets high resolution, reaches 12.0% at its fixed 1536 × 1536 input. Tile-based high-resolution methods [16, 17] sharpen detail by fusing crops with a low-resolution global estimate from which they inherit the metric scale; we expect them to sharpen edges rather than remove the scale error, but we have not evaluated them.

Licences. Licences differ more than accuracy might suggest (table 1). libSGM (Apache-2.0), Depth Anything V2-S and Depth Anything 3 Metric-Large (Apache-2.0), MoGe-2 (MIT) and Metric3D v2 (BSD-2-Clause) are permissive; ESS is distributed under NVIDIA's model EULA, whose model card states that it is ready for commercial use; C-Fast-FoundationStereo is governed by the NVIDIA Open Model Agreement, although its model card also describes it as intended for research and evaluation; the Fast-FoundationStereo research checkpoints (NVIDIA Source Code License), Depth Anything V2-B/L and UniDepth V2 (CC BY-NC 4.0) are licensed for non-commercial use only, and Depth Pro under Apple's own licence. We report the licences as the authors state them; they are not legal advice.

Coverage. The benchmark covers the model families we consider most relevant for metric depth on an Orin as of October 2026, not every published method. We did not evaluate the Orin's optical flow accelerator through VPI, FoundationStereo itself (Fast-FoundationStereo is its real-time version), other learned stereo networks such as RAFT-Stereo, or tile-based high-resolution monocular methods.

Edges and the limits of AbsRel. The mean AbsRel over predicted pixels rewards dense, smooth output and is nearly blind to depth discontinuities (section 6.3). It also treats a missing pixel as neutral, which favours sparse stereo, and a flying point as a small error, which favours the networks. Neither effect changes the ranking between stereo and monocular methods here, but it means that the most accurate method by AbsRel is not necessarily the best at object boundaries: a classical matcher at full resolution produces the cleanest boundaries of all methods we tested.

Scope of the data. Our 10 Middlebury 2014 scenes are indoor and close-range (1.3–7.6 m), with 17–24 cm

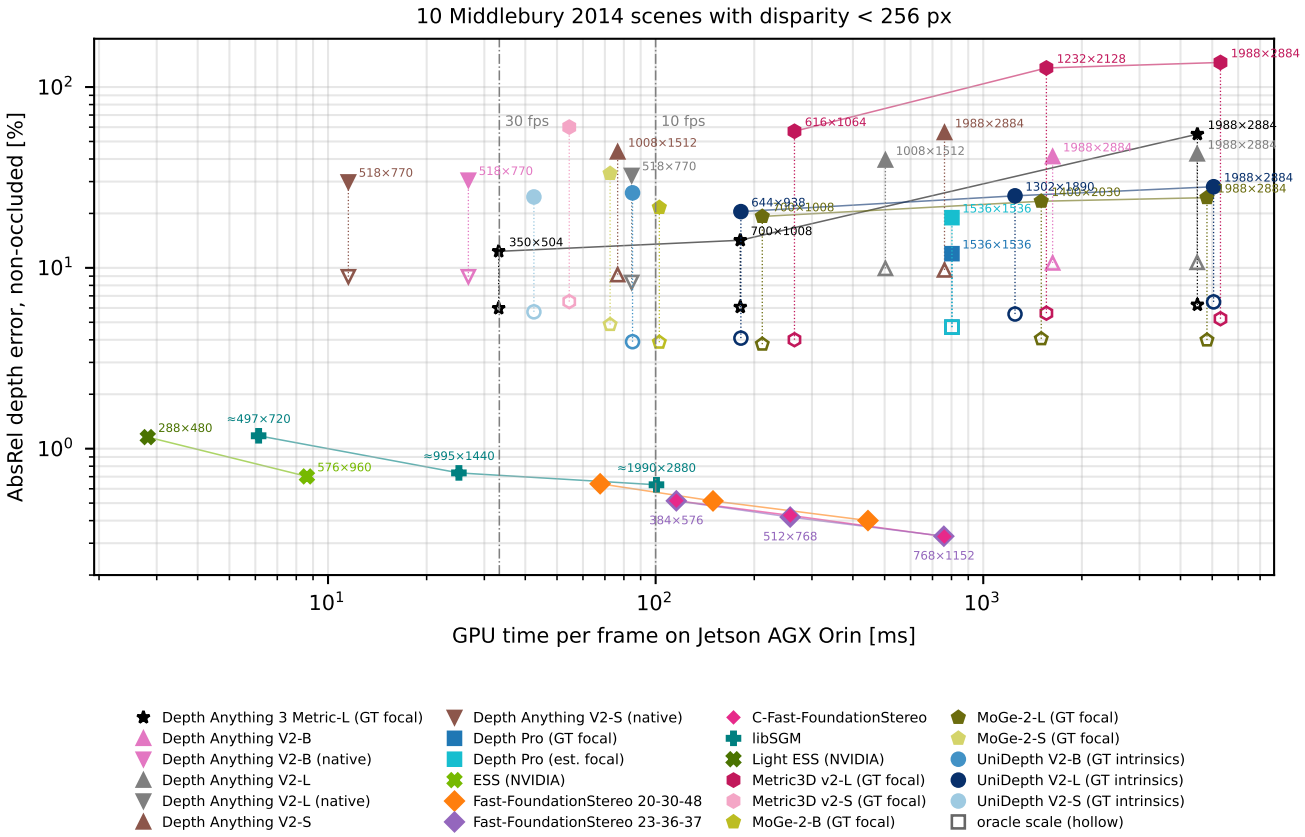


Figure 5: Depth error (AbsRel on non-occluded pixels, log scale) against GPU time per frame on Jetson AGX Orin, averaged over the 10 scenes. Filled markers are the methods as deployed (raw output; hole-filled results are in table 7 only); hollow markers are monocular methods with an oracle per-image scale. Upward triangles are Depth Anything V2 at full resolution, downward triangles at its native 518×770 input. The black stars are Depth Anything 3 Metric-L and the olive pentagons MoGe-2-S/B/L, the pink hexagons Metric3D v2-S/L and the blue circles UniDepth V2-S/B/L, with the ground-truth focal length or intrinsics; thin lines connect the native, twice-native and full-resolution runs of Depth Anything 3, MoGe-2-L, Metric3D v2-L and UniDepth V2-L (the smaller models at other sizes and MoGe-2 with its own focal estimate are in table 7 only). Green crosses are ESS and Light ESS, connected by a thin line; thin lines also connect the three input resolutions of libSGM (teal crosses) and of each Fast-FoundationStereo model. The small pink diamonds (C-Fast-FoundationStereo) coincide with the purple ones of the research checkpoint `23-36-37`. Labels give the processing resolution (height×width). Vertical lines mark the per-frame budgets for 30 and 10 fps on one Orin.

baselines and focal lengths of 2,800–7,200 px. Disparities are therefore large and stereo triangulation is very accurate. Stereo depth error grows quadratically with range, so at long range with a short baseline the gap would shrink; wide-baseline automotive stereo is designed to counter this. A long-range, outdoor benchmark with LiDAR ground truth is the most important follow-up.

Fairness to monocular methods. We gave the monocular models the true focal length or intrinsics (Depth Pro, Depth Anything 3, MoGe-2, Metric3D v2, UniDepth V2), a domain-matched indoor checkpoint (Depth Anything V2 Hypersim) and an oracle scale, none of which is available in deployment. For MoGe-2 the true focal length helps the shape (lower oracle-scaled error) but not the metric scale, which comes from a separate head. Evaluating on non-occluded pixels removes the 9% of the ground truth that stereo cannot triangulate by construction; the restriction favours stereo only by excluding those pixels, which the hole-filled rows of table 7 account for. We did not fine-tune any model on Middlebury. We cannot rule out that Middlebury images appear in any model's training data; if they do, that would favour the networks. A model fine-tuned for a specific camera would close part of the raw-to-oracle gap but not the oracle gap itself.

Disparity range and scene selection. libSGM searches at most 256 disparity levels at its processing resolution, which at full resolution covers 255 px. We there-

Table 5: System cost: cameras plus the fraction of Jetson AGX Orin 64GB GPU time used ($2,999 per module). Stereo uses two cameras and monocular one. "Orins" above 1 require multiple modules (or a larger GPU) for that frame rate. Break-even: the camera price below which adding the second camera and running Light ESS, the cheapest stereo configuration, is cheaper than the monocular model (eq. (1)). Monocular models with their own focal-length estimate have the same cost as with the true one and are omitted.

Method	Resolution	Orins @10 fps	Orins @30 fps	$@30 fps, $20 per camera	$@30 fps, $500 per camera	Break-even camera @30 fps
Light ESS (NVIDIA)	288×480 (fixed)	0.03	0.08	293	1,253	–
libSGM	≈497×720 (quarter)	0.06	0.18	592	1,552	–
ESS (NVIDIA)	576×960 (fixed)	0.09	0.26	816	1,776	–
libSGM	≈995×1440 (half)	0.25	0.75	2,297	3,257	–
Fast-FoundationStereo 20-30-48	384×576	0.68	2.03	6,132	7,092	–
libSGM	full (≈1990×2880)	1.01	3.02	9,090	10,050	–
Fast-FoundationStereo 23-36-37	384×576	1.16	3.47	10,445	11,405	–
C-Fast-FoundationStereo	384×576	1.16	3.47	10,446	11,406	–
Fast-FoundationStereo 20-30-48	512×768	1.50	4.49	13,505	14,465	–
Fast-FoundationStereo 23-36-37	512×768	2.57	7.72	23,204	24,164	–
C-Fast-FoundationStereo	512×768	2.58	7.73	23,218	24,178	–
Fast-FoundationStereo 20-30-48	768×1152 (max)	4.45	13.34	40,054	41,014	–
Fast-FoundationStereo 23-36-37	768×1152 (max)	7.58	22.74	68,247	69,207	–
C-Fast-FoundationStereo	768×1152 (max)	7.60	22.81	68,454	69,414	–
Depth Anything V2-S (native)	518×770 (native)	0.12	0.35	1,058	1,538	$784
Depth Anything V2-B (native)	518×770 (native)	0.27	0.80	2,433	2,913	$2,159
Depth Anything 3 Metric-L (GT focal)	350×504 (native)	0.33	1.00	3,005	3,485	$2,732
UniDepth V2-S (GT intrinsics)	644×938 (native)	0.43	1.28	3,844	4,324	$3,571
Metric3D v2-S (GT focal)	616×1064 (native)	0.54	1.63	4,922	5,402	$4,649
MoGe-2-S (GT focal)	700×1008 (native)	0.73	2.18	6,553	7,033	$6,279
Depth Anything V2-S	1008×1512	0.77	2.30	6,908	7,388	$6,635
Depth Anything V2-L (native)	518×770 (native)	0.84	2.53	7,621	8,101	$7,348
UniDepth V2-B (GT intrinsics)	644×938 (native)	0.85	2.55	7,669	8,149	$7,395
MoGe-2-B (GT focal)	700×1008 (native)	1.03	3.08	9,253	9,733	$8,980
Depth Anything 3 Metric-L (GT focal)	700×1008	1.81	5.44	16,330	16,810	$16,056
UniDepth V2-L (GT intrinsics)	644×938 (native)	1.82	5.46	16,400	16,880	$16,126
MoGe-2-L (GT focal)	700×1008 (native)	2.12	6.35	19,056	19,536	$18,783
Metric3D v2-L (GT focal)	616×1064 (native)	2.65	7.96	23,890	24,370	$23,617
UniDepth V2-S (GT intrinsics)	1302×1890	2.79	8.37	25,133	25,613	$24,860
Metric3D v2-S (GT focal)	1232×2128	2.96	8.87	26,612	27,092	$26,339
MoGe-2-S (GT focal)	1400×2030	4.19	12.56	37,695	38,175	$37,421
Depth Anything V2-L	1008×1512	5.03	15.08	45,251	45,731	$44,977
UniDepth V2-B (GT intrinsics)	1302×1890	5.80	17.39	52,169	52,649	$51,896
MoGe-2-B (GT focal)	1400×2030	6.63	19.90	59,710	60,190	$59,437
Depth Anything V2-S	1988×2884 (full)	7.62	22.85	68,539	69,019	$68,265
Depth Pro (GT focal)	1536×1536 (fixed)	8.03	24.08	72,221	72,701	$71,948
Metric3D v2-S (GT focal)	1988×2884 (full)	9.92	29.77	89,301	89,781	$89,028
UniDepth V2-S (GT intrinsics)	1988×2884 (full)	11.33	34.00	101,987	102,467	$101,713
MoGe-2-S (GT focal)	1988×2884 (full)	11.79	35.36	106,062	106,542	$105,789
UniDepth V2-L (GT intrinsics)	1302×1890	12.50	37.50	112,492	112,972	$112,219
MoGe-2-L (GT focal)	1400×2030	15.03	45.10	135,274	135,754	$135,000
Metric3D v2-L (GT focal)	1232×2128	15.59	46.76	140,244	140,724	$139,970
Depth Anything V2-B	1988×2884 (full)	16.28	48.85	146,528	147,008	$146,255
MoGe-2-B (GT focal)	1988×2884 (full)	20.07	60.21	180,591	181,071	$180,317
UniDepth V2-B (GT intrinsics)	1988×2884 (full)	23.35	70.06	210,120	210,600	$209,846
Depth Anything V2-L	1988×2884 (full)	44.97	134.91	404,605	405,085	$404,332
Depth Anything 3 Metric-L (GT focal)	1988×2884 (full)	45.04	135.11	405,218	405,698	$404,944
MoGe-2-L (GT focal)	1988×2884 (full)	48.18	144.54	433,486	433,966	$433,213
UniDepth V2-L (GT intrinsics)	1988×2884 (full)	50.48	151.43	454,144	454,624	$453,870
Metric3D v2-L (GT focal)	1988×2884 (full)	52.98	158.93	476,638	477,118	$476,365

fore restrict the evaluation to the 10 scenes whose non-occluded disparity stays below that range; in the other 13 public scenes, near objects at 265–712 px of disparity would be out of range for libSGM at full resolution, which would turn a search-range setting into a large accuracy penalty. This selection favours stereo only in the sense that it removes a configuration problem with a known fix: running the matcher at lower resolution (at half resolution 256 levels span ≈510 px, at quarter resolution ≈1020 px), or a matcher with a larger search range. It also removes the closest-range scenes, in which stereo depth is most accurate.

Parameter tuning. libSGM's P_1 and P_2 were tuned on the evaluation scenes, which favours it; cross-validation with one scene held out raises its AbsRel by only 0.01–0.08 percentage points (appendix C), so the conclusions do not depend on it. The networks use their released weights without any tuning.

Timing methodology. Network timings exclude image resizing, normalisation and post-processing (for MoGe-2 the focal-length and shift recovery and the metric-scale head), while the SGM timings include copying the images to and the disparity from the GPU, so the comparison slightly favours the networks. The fractional cost model assumes compute can be shared across tasks; with dedicated hardware, cost rises in steps of whole modules.

9 Conclusion

On a Jetson AGX Orin, stereo gives metric depth that no current monocular foundation model approaches. At 30 fps, NVIDIA's ESS reaches 0.70% AbsRel with dense

output in 8.6 ms and libSGM 0.73% in 25 ms; with more compute Fast-FoundationStereo reaches 0.33%, as does its commercially licensed checkpoint. The best monocular models, given the true focal length, reach 12% (Depth Anything 3 Metric-L in 33 ms, Depth Pro in 803 ms), and only with an oracle per-image scale do MoGe-2-L, UniDepth V2-B and Metric3D v2-L reach 3.8–4.0%, still several times the error of stereo. Stereo error falls as the input resolution grows, while no monocular model improves, because its metric scale drifts outside the training resolution and its shape does not sharpen; and at object boundaries the monocular models scatter 17–46% of the points between surfaces, an error that AbsRel barely registers and where a classical matcher at full resolution is the cleanest of all. For metric depth on an Orin, a second camera with ESS, libSGM or Fast-FoundationStereo is the better choice at every compute budget we measured; monocular models are the choice when only one camera is possible and the metric scale can be recovered by other means.

Conflict of interest

The author is employed by NODAR Inc., which develops and sells stereo vision products, and has a financial interest in stereo vision. For fairness, we did not include NODAR's own stereo matcher, Hammerhead, in the comparison: no NODAR product is evaluated here, and every method is open source or publicly released and is run with publicly documented settings on the same hardware and the same public ground truth. All evaluation code, per-scene results and engine build scripts are released, and the monocular methods are additionally given the true focal length and an oracle scale.

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A Dataset

Table 6 lists the 10 Middlebury 2014 scenes [10] that we use: the scenes with public ground truth whose non-occluded disparity stays below 256 px, the range of the 256-level semi-global matcher at full resolution. We use them at full resolution with "perfect" rectification. Ground-truth disparity comes from structured light and is given in full-resolution pixels; `ndisp` is the search bound in Middlebury's calibration file, not the true maximum. *Non-occ. max* is the largest disparity among pixels visible in both views, which is the ground truth we evaluate. *Occl.* and *Out* are the fractions of ground-truth pixels occluded in the right view or outside its field of view. The other 13 public scenes (Cable, Couch, Flowers, Jadeplant, Mask, Pipes, Playroom, Playtable, Shopvac, Sticks, Storage, Sword2, Vintage) have non-occluded disparities of 265–712 px and are not used.

Table 6: The 10 Middlebury 2014 scenes used (left image thumbnails). Disparities in full-resolution pixels; depth from $Z = fB/(d + d_{\text{offs}})$.

				GT disparity [px]				Non-occ.	Depth	Occl.	Out
Image	Scene	Size ($W \times H$)	`ndisp`	min	median	99 %	max	max [px]	[m]	[%]	[%]
	Adirondack	2880×1988	280	30	108	219	243	243	1.62–3.07	6.1	2.0
	Backpack	2940×2016	260	32	130	214	224	224	2.22–3.36	8.4	3.1
	Bicycle1	2988×2008	180	56	73	127	145	145	2.94–4.08	6.3	2.6
	Classroom1	3000×1920	250	16	55	203	208	208	2.98–7.63	2.7	1.8
	Motorcycle	2964×2000	270	28	152	231	240	240	2.11–5.04	7.9	3.1
	Piano	2820×1920	260	38	104	184	209	209	1.51–3.11	5.0	3.6
	Recycle	2864×1924	260	37	102	203	220	220	1.34–2.53	5.2	2.5
	Shelves	2952×2000	240	54	113	192	200	200	2.53–4.14	4.2	2.8
	Sword1	2912×2020	260	5	95	204	233	233	2.28–4.02	7.3	3.0
	Umbrella	2960×2016	250	43	113	210	218	212	3.04–6.44	5.1	2.5

B Full results

Table 7: Accuracy (non-occluded ground truth only) and cost per frame on Jetson AGX Orin (MAXN). Dens.: fraction of non-occluded ground-truth pixels with a prediction. bad-2 counts missing predictions as errors; AbsRel and $\delta_{1.05}$ are over predicted pixels. AbsRel†: monocular output after per-image oracle median scaling. GPU ms: median TensorRT GPU compute time for networks and median Nsight Systems GPU-busy time for libSGM. J/frame: dynamic SoC energy. Accuracy of Depth Pro, Depth Anything V2-S at full resolution, Depth Anything 3, MoGe-2, Metric3D v2 and UniDepth V2 is from FP32 PyTorch and their timing from the FP16 engines (section 6.5).

Method	Resolution	Dens. [%]	bad-2 [%]	AbsRel [%]	$\delta_{1.05}$ [%]	AbsRel† [%]	GPU ms	J/frame
Light ESS (NVIDIA)	288×480 (fixed)	100	36.9	1.16	96.7	–	2.8	0.08
libSGM	≈497×720 (quarter)	88	31.3	1.18	94.6	–	6.1	0.18
libSGM (hole-filled)	≈497×720 (quarter)	100	28.4	2.13	90.3	–	6.1	0.18
ESS (NVIDIA)	576×960 (fixed)	100	20.4	0.70	98.4	–	8.6	0.31
libSGM	≈995×1440 (half)	85	24.7	0.73	96.8	–	25.1	0.73
libSGM (hole-filled)	≈995×1440 (half)	100	20.2	1.79	92.1	–	25.1	0.73
Fast-FoundationStereo 20-30-48	384×576	100	17.0	0.64	98.2	–	67.7	1.88
libSGM	full (≈1990×2880)	76	32.1	0.63	97.5	–	100.6	2.96
libSGM (hole-filled)	full (≈1990×2880)	100	22.3	1.92	91.4	–	100.6	2.96
Fast-FoundationStereo 23-36-37	384×576	100	12.7	0.51	98.7	–	115.6	3.32
C-Fast-FoundationStereo	384×576	100	12.9	0.52	98.7	–	115.7	3.29
Fast-FoundationStereo 20-30-48	512×768	100	12.4	0.51	98.7	–	149.7	4.23
Fast-FoundationStereo 23-36-37	512×768	100	8.9	0.42	99.0	–	257.5	7.43
C-Fast-FoundationStereo	512×768	100	9.4	0.43	99.0	–	257.6	7.39
Fast-FoundationStereo 20-30-48	768×1152 (max)	100	8.5	0.40	99.0	–	444.7	12.93
Fast-FoundationStereo 23-36-37	768×1152 (max)	100	6.5	0.33	99.3	–	758.1	22.39
C-Fast-FoundationStereo	768×1152 (max)	100	6.0	0.33	99.3	–	760.4	21.95
Depth Anything V2-S (native)	518×770 (native)	100	98.1	29.88	11.0	8.93	11.5	0.50
Depth Anything V2-B (native)	518×770 (native)	100	98.5	30.55	7.3	8.93	26.8	1.32
Depth Anything 3 Metric-L (GT focal)	350×504 (native)	93	95.0	12.34	30.8	5.97	33.2	1.69
UniDepth V2-S (GT intrinsics)	644×938 (native)	93	98.4	24.67	11.9	5.71	42.5	1.72
Metric3D v2-S (GT focal)	616×1064 (native)	65	98.8	60.06	5.5	6.51	54.5	2.01
MoGe-2-S (GT focal)	700×1008 (native)	76	98.5	33.26	7.5	4.86	72.6	2.77
MoGe-2-S (est. focal)	700×1008 (native)	100	97.7	30.33	10.7	9.06	72.6	2.77
Depth Anything V2-S	1008×1512	100	99.9	43.79	0.5	9.17	76.6	3.40
Depth Anything V2-L (native)	518×770 (native)	100	98.5	32.46	9.3	8.31	84.5	4.12
UniDepth V2-B (GT intrinsics)	644×938 (native)	84	98.6	25.99	11.6	3.90	85.0	3.69
MoGe-2-B (GT focal)	700×1008 (native)	96	97.5	21.58	17.5	3.88	102.6	4.19
MoGe-2-B (est. focal)	700×1008 (native)	100	99.7	36.41	1.5	7.53	102.6	4.19
Depth Anything 3 Metric-L (GT focal)	700×1008	90	96.8	14.21	23.0	6.07	181.3	8.56
UniDepth V2-L (GT intrinsics)	644×938 (native)	99	96.4	20.47	12.5	4.09	182.1	8.28
MoGe-2-L (GT focal)	700×1008 (native)	94	95.8	19.24	13.0	3.78	211.6	9.28
MoGe-2-L (est. focal)	700×1008 (native)	100	98.6	39.87	6.9	9.01	211.6	9.28
Metric3D v2-L (GT focal)	616×1064 (native)	63	100.0	56.93	0.0	4.00	265.3	11.97
UniDepth V2-S (GT intrinsics)	1302×1890	95	97.5	23.37	10.2	8.02	279.1	12.22
Metric3D v2-S (GT focal)	1232×2128	14	100.0	133.00	0.0	7.94	295.6	12.11
MoGe-2-S (GT focal)	1400×2030	68	98.5	42.94	7.2	5.60	418.7	16.71
MoGe-2-S (est. focal)	1400×2030	100	99.8	36.67	1.7	9.75	418.7	16.71
Depth Anything V2-L	1008×1512	100	99.6	39.46	3.6	9.92	502.7	23.54
UniDepth V2-B (GT intrinsics)	1302×1890	95	96.8	21.21	19.0	5.83	579.6	25.47
MoGe-2-B (GT focal)	1400×2030	86	99.1	34.36	8.2	4.43	663.4	28.11
MoGe-2-B (est. focal)	1400×2030	100	99.7	37.07	4.2	8.26	663.4	28.11
Depth Anything V2-S	1988×2884 (full)	100	100.0	56.04	0.1	9.73	761.6	31.70
Depth Pro (GT focal)	1536×1536 (fixed)	98	95.9	11.98	28.9	4.70	802.5	38.68
Depth Pro (est. focal)	1536×1536 (fixed)	95	99.4	18.95	10.5	4.70	802.5	38.68
Metric3D v2-S (GT focal)	1988×2884 (full)	5	100.0	162.46	0.0	8.44	992.3	41.69
UniDepth V2-S (GT intrinsics)	1988×2884 (full)	96	98.2	22.57	10.1	8.96	1,133.3	50.43
MoGe-2-S (GT focal)	1988×2884 (full)	66	94.8	40.23	12.5	6.00	1,178.6	48.76
MoGe-2-S (est. focal)	1988×2884 (full)	100	99.3	41.33	2.6	9.99	1,178.6	48.76
UniDepth V2-L (GT intrinsics)	1302×1890	100	98.3	25.02	8.3	5.55	1,250.1	56.50
MoGe-2-L (GT focal)	1400×2030	87	97.1	23.36	13.9	4.05	1,503.3	67.23
MoGe-2-L (est. focal)	1400×2030	100	99.1	40.65	5.1	9.33	1,503.3	67.23
Metric3D v2-L (GT focal)	1232×2128	9	100.0	127.52	0.0	5.61	1,558.6	70.16
Depth Anything V2-B	1988×2884 (full)	100	100.0	41.24	0.5	10.64	1,628.4	72.17
MoGe-2-B (GT focal)	1988×2884 (full)	81	99.0	37.23	5.3	4.54	2,007.0	86.64
MoGe-2-B (est. focal)	1988×2884 (full)	100	99.6	36.32	3.3	8.35	2,007.0	86.64
UniDepth V2-B (GT intrinsics)	1988×2884 (full)	96	99.4	26.18	8.4	7.69	2,335.2	103.84
Depth Anything V2-L	1988×2884 (full)	100	99.7	42.69	2.3	10.72	4,496.9	202.70
Depth Anything 3 Metric-L (GT focal)	1988×2884 (full)	56	100.0	54.88	0.2	6.24	4,503.7	201.72
MoGe-2-L (GT focal)	1988×2884 (full)	86	99.1	24.45	13.1	3.99	4,817.9	214.81
MoGe-2-L (est. focal)	1988×2884 (full)	100	98.6	39.67	7.7	9.17	4,817.9	214.81

C libSGM parameters

Table 8 lists the libSGM parameters we use. P_1 and P_2 were chosen separately for each input resolution by a grid search ($P_1 \in \{1, 2, 3, 4, 6, 10, 15, 20, 30\}$, $P_2 \in \{40, 60, 90, 120, 160, 200, 224\}$, $P_1 < P_2$) that minimises the mean AbsRel over the 10 evaluation scenes, scored exactly as in table 7. All other parameters are libSGM's defaults, except that we use four aggregation paths instead of eight and enable sub-pixel output. libSGM stores aggregated path costs in 8 bits, so the census cost (at most 31 for the symmetric 9×7 census) plus P_2 must stay below 256; the largest usable P_2 is therefore 224, and the search selected it at every resolution.

The search has no separate training set, so the selected parameters are tuned on the evaluation scenes, which favours libSGM. To estimate how much, we repeated the selection ten times with one scene held out and scored each held-out scene with the parameters chosen on the other nine. The held-out mean AbsRel is 0.65%, 0.75% and 1.26% at full, half and quarter resolution, against 0.63%, 0.73% and 1.18% when tuned on all ten scenes and 1.03%, 1.04% and 1.29% with the default $P_1 = 10$, $P_2 = 120$. The held-out choices agree with the selected ones in 9 of 10 folds at full and half resolution and in 7 of 10 at quarter resolution. Minimising AbsRel over predicted pixels also lowers the density: the stronger smoothing makes more pixels fail the left–right check (76%, 85% and 88% density, against 84%, 89% and 91% with the defaults).

Table 8: libSGM (version 3.1.0, commit `e4c669b`) parameters.

Parameter	Value	Note
P_1	4 (full), 6 (half), 3 (quarter)	tuned; default 10
P_2	224 at every resolution	tuned; default 120; 224 is the 8-bit maximum
Aggregation paths	4 (horizontal and vertical)	default 8
Sub-pixel output	on (4 fractional bits)	default off
Disparity levels	256	maximum supported
Minimum disparity	0	default
Census	symmetric 9×7 (31 bits)	default
Uniqueness	0.95	default
Left–right check tolerance	1 px	default
Median filter	3×3	always applied
Input	8-bit grey, area-downsampled for half and quarter	
Output	16-bit, host-to-host (copies included in timing)	